Fiscal Year:	FY 2015	**Task Last Updated:**	FY 08/12/2014
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Individual Differences in the Neurochemical and Behavioral Response to Exposure to Protons and HZE Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Shukitt-Hale, Barbara (US Department of Agriculture/Agricultural Research Service, Beltsville)		
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Task Description:	Future missions in space (such as a mission to Mars or the establishment of a moon base) will involve long-term travel beyond the magnetic field of the Earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. Cosmic rays are composed of protons, alpha particles, and particles of high energy and charge (HZE particles). Previous research has shown that exposure to HZE particles and protons can affect neural functioning and behavior. These changes may affect the ability of astronauts to successfully meet mission requirements or may produce degenerative changes in neural and behavioral functioning after the conclusion of the mission. The objectives of the proposed research program are: (1) to evaluate the effects of repeated subthreshold exposures to HZE particles and protons on cognitive and neuronal deficits and how they relate to the characteristics of the specific particle; (2) to determine the relationship between changes in performance and HZE particle- and proton-induced changes in oxidative stress, inflammation, signaling molecules, and autophagy in key brain regions; and (3) to determine how individual		

	characteristics, including age and sex, may influence the responsiveness of the individual to HZE particle irradiation. To achieve these objectives, we propose using behavioral and neurochemical measures that have been previously shown to be sensitive to exposure to HZE particles and protons and which may provide a basis for defining the risks of exposure to heavy particles on brain functioning and related behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Two sets of experiments have been performed. The first experiment evaluated the acute effects (24 hr) of exposure to 16O or 56Fe particles on learning and on memory. The results of this experiment testing for the acute effects of exposure to HZE particles showed that exposing the rats to either 56Fe or 16O particles after conditioning (learning) but before testing, disrupted recognition memory: the rats spent equal amounts time with the familiar and novel objects, indicating that they did not recognize the familiar object. In contrast to the test of recognition memory, there was no effect of exposure to either particle on the ability of the subjects to learn the task when conditioning occurred 24 hr after exposure. When the rats were irradiated prior to conditioning, subjects were able to recognize the familiar object, as shown by the greater amount of time spent with the novel object in comparison to the familiar object. The results indicate that there are acute effects on cognitive performance. The disruption of memory by interposing exposure to HZE particles between learning (conditioning) and testing suggests that the irradiation interferes with the consolidation process: the process by which a memory is transferred from short-term to long-term memory. In the learning paradigm, in contrast, irradiation occurs before conditioning (learning) and therefore does not affect the consolidation process and recognition memory is preserved. Neurochemical analyses of hippocampus, frontal cortex, striatum, and cortex showed significant differences between the irradiated and control animals for both the OS marker NOX2 and the inflammatory marker COX2. Overall radiation effects were significant between control and radiated animals for 56Fe and 16O. Exposure to 16O particles produced significant differences between memory and learning paradigms in the frontal cortex, striatum, cortex, and hippocampus regions for the levels of NOX2. However, for the 56Fe exposure, the significant differences between memory and learning paradigms was observed in frontal cortex and cortex region only. Both 56Fe and 16O particle exposure affected the frontal cortex which plays a critical role in the working memory and memory consolidation. The results indicate acute exposure to HZE caused an acute overload of ROS in the key memory consolidation region of the brain in the memory paradigm but not in the learning set of animals. Similar to NOX2 which can generate oxidative reactive oxygen species that can impair memory, levels of COX2, a pro-inflammatory mediator, were also significantly affected between memory and learning paradigms in the hippocampus when animals were exposed to acute doses 56Fe or 16O. Significant differences between memory and learning for the expression of COX-2 were also observed in the frontal cortex, striatum, and cerebellum when the animals were exposed to 16O. The second series of experiments evaluated the effects of repeated sub-threshold exposure to 56Fe. Most research on the cognitive effects of exposure to the types of radiation encountered in space has used a single suprathreshold dose of HZE particles or protons. However, on exploratory class missions it most likely that the total dose received by astronauts will be spread across the duration of the mission. Performance was evaluated on the plus-maze (baseline anxiety), novel spatial recognition (spatial learning and memory), novel object recognition (non-spatial learning and memory), and operant responding (responsiveness to changes in reinforcement contingencies). The radiations were single exposures to 15 or 30 cGy, or two exposures to 15 cGy separated by either 24 or 72 hr. The results suggest that the effects of multiple exposures to 56Fe particles are task-dependent. Plus-maze performance showed no effects following single-dose exposure to either 15 or 30 cGy. However, there was a significant increase in the amount of time spent in the anxiogenic open arms of the maze following two exposures to 15 cGy of 56Fe particles separated by 24 or 72 hours. A similar pattern was observed for novel spatial recognition. In contrast, performance on the novel object recognition task and on the operant task was disrupted only by a single dose of 30 cGy. Other than noting that the effects of multiple exposures apparently are task-dependent, the factors that affect responding under these exposures conditions are not known. We are currently in the process of retesting the subjects on these tasks to evaluate the effects of aging on performance. Also, because the different tasks are dependent upon the integrity of different brain areas, it possible that the differences in performance reflect the different sensitivities of these areas to HZE particle-induced oxidative stress. Plans are underway to evaluate the role of HZE particle-induced oxidative stress and neuroinflammation as they affect the performance of organisms to repeated exposure to subthreshold doses of HZE particles.
Bibliography Type:	Description: (Last Updated: 05/08/2025)
Articles in Peer-reviewed Journals	Rabin BM, Shukitt-Hale B, Carrihill-Knoll KL. "Effects of age on the disruption of cognitive performance by exposure to space radiation." Journal of Behavioral and Brain Science. 2014 Jul;4(7):297-307. http://dx.doi.org/10.4236/jbbs.2014.47031 , Jul-2014
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Articles in Peer-reviewed Journals	Rabin BM, Shukitt-Hale B. "A voyage to Mars: Space radiation, aging, and nutrition." Nutrition and Aging. 2014;2(4):233-41. http://dx.doi.org/10.3233/NUA-140045 , Aug-2014
Articles in Peer-reviewed Journals	Rabin BM, Shukitt-Hale B, Gomes S, Carrihill-Knoll, KL. "Cognitive effects of partial and whole body exposures to 16O particles." J Radiat Res. 2014 Mar;55(Suppl 1):i100-i101. (Proceedings of Heavy Ion in Therapy and Space Radiation Symposium 2013/6th International Workshop on Space Radiation Research, Chiba, Japan, May 15-18, 2013.) http://dx.doi.org/10.1093/jrr/rrt187 , Mar-2014
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